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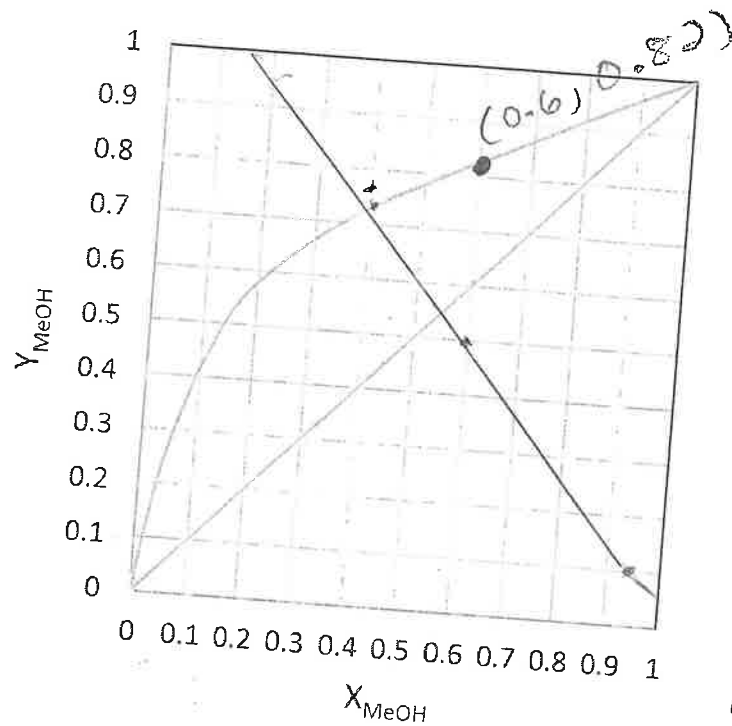
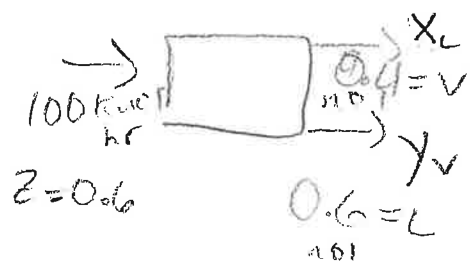
CHE 313 Transport Phenomena III

Quiz #1 (9/16/25) **Name:** Diata Rodriguez

1. We are separating a mixture of methanol and water in a flash drum at 1.0 atm pressure. Feed is 60.0 mol% methanol, and 40.0 % of the feed is vaporized. Feed flow rate is 100 kmol/hr.

   Equilibrium xy graph is:





F = Lx + Vy

Fz = Lx + Vy

a) (2pt) Sketch the process, define all variables, and set up overall mass balance and component mass balance equations.

b) (2pts) Derive an operating line equation using component mass balance equations (don't just write down the operating equation).

c) (2pts) Find liquid flow rate (L), vapor flow rate (V), a slope of the operating line, and a point on y=x curve that operating line should pass.

d) (2pts) Draw the operating line on the graph, and find the vapor (y) and liquid (x) mole fractions.

(0.41, 0.72)

e) (2pts) By changing the amount of evaporated mixture, we want to achieve the maximum possible methanol content in the distillate. Use graphical solution method.

vapor

y = z

overall: $F = L + V$

a) $Fz = Lx + Vy$

component

$Fz_i\ Lx_i + Vy_i$

2

$\frac{100\,kmol}{hr}$

$z = 0.6$

$V = 0.40$ y_M

40 mol %

Flow rate

40 % feed vaporized

$L = 60$ mol % x_M

Means 60 % is Liquid

c) $y = -\frac{L x_i}{V} + \frac{F z_i}{V}$

$y = -\frac{160}{401} x + \frac{100\,\frac{kmol}{hr}}{40} (0.6)$

1 $y = -1.5x + 1.5$

$y = -\frac{3}{2}x + 3/2$

$L = 60\,\frac{kmol}{hr}$ $V = 40\,\frac{kmol}{hr}$

slope $= -1.5$

b) $Fz = Lx + Vy$

$Vy = -Lx + Fz$ 2

$y = -\frac{L}{V}x_i + \frac{F}{V}z_i$

(Do we find $\frac{V}{F}$?)

point on $y = x$..

$(0.6, 0.6)$

$\frac{V}{F} = f$ $-\frac{L}{V}$ $\frac{2}{5}$ $\frac{3}{5}$ $\frac{\frac{3}{5}}{2/5}$

d) where operating line crosses vapor line this would give x & y

x & y values well that specific coordinate would be x & y mol fractions

$(0.48, 0.78)$

1

DiATA Rodrigue?

e) $x = 0.6$

$y = 0.82$ 2

Maximum possible methanol content is when $x = 0.6$